

Case Study 2: Exposé

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Evaluating EU AI Act Compliance Assessment Agents:

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Abstract

The EU AI Act is now in force, and one of its first practical demands is deciding whether a given AI system counts as high risk under Article 6. This decision is slow and legally delicate, so organisations are starting to build language model agents that read a system description and return a classification. The usual way to check whether such an agent can be trusted is an automated faithfulness metric, which measures whether the agent's answer is grounded in the text it retrieved. This case study asks whether faithfulness alone is enough. An answer can be perfectly grounded in the retrieved guidelines and still reach the wrong legal classification, and the faithfulness metric cannot see that gap because it never looks at the correct answer. To show this, I build one retrieval based compliance agent as a test specimen and run it over up to 150 system descriptions whose correct high risk labels are fixed in advance from the European Commission's May 2026 draft guidelines. Each answer is measured on two independent axes: whether the legal label is correct, and whether the answer is faithful to the retrieved text. The contribution is to test whether these two axes come apart, whether a well grounded answer can be legally wrong, and whether the standard metric misses it. The result speaks directly to how reliable AI generated compliance assessments really are, and to what it means for the role of human oversight in the legal check.

1. Introduction

1.1 Motivation

The EU AI Act is the first broad law that regulates artificial intelligence according to how risky a system is. One of its central steps is Article 6, which decides whether a system is high risk. That label matters a great deal, because a high risk system carries a long list of legal obligations while a system outside that category carries almost none. Getting the classification right is therefore the gate that everything else in a compliance process depends on.

Making this call by hand is slow and needs careful legal reading, so it is a natural target for automation. A common approach is a retrieval based agent, often called a RAG system, which stands for retrieval augmented generation. Retrieval augmented generation means the system first searches a body of text for the passages most relevant to a question, then feeds those passages to a language model so the model answers from real source text rather than only from its own memory. Point such an agent at the AI Act guidance, give it a description of a system, and it returns a classification with an explanation drawn from that guidance. On the surface this looks like a solved problem.

1.2 Challenges and Problem Statement

The real problem is how we decide whether to trust the agent's answer. The field standard for checking a retrieval based system is a faithfulness metric. Faithfulness measures one thing only: whether the claims in the answer are supported by the passages the system retrieved. It is a grounding check. It does not, and by design cannot, check whether the final legal classification is actually correct, because it never sees the correct answer.

This creates a blind spot that matters for law. An agent can retrieve exactly the right passage, reason from it in a way that is fully grounded and internally consistent, and still land on the wrong classification, for example by misapplying the Article 6(3) exception that can pull a system back out of the high risk category. The faithfulness metric would score that answer highly, because every claim traces back to real retrieved text. The answer is faithful and wrong at the same time, and the metric that everyone relies on would give it a green light. The reverse can also happen: the agent reaches the correct label while leaning on text that does not really support it, so the answer is right but its reasoning is not trustworthy. In both cases the metric and the legal truth disagree, and the metric cannot tell.

1.3 Contribution and Research Question

This case study puts that blind spot on trial. I build one retrieval based compliance agent as a test specimen, not as a product, and I evaluate its answers on two separate axes that are usually collapsed into one. The first axis is correctness, meaning whether the legal label is right, checked against fixed ground truth. The second axis is faithfulness, meaning whether the answer is grounded in the retrieved text, measured with the standard automated metric. The contribution is to test whether these two axes are independent, whether a faithful answer can be legally wrong and a correct answer can be unfaithful, and whether the standard metric can detect either failure.

Research question. To what extent do automated faithfulness metrics fail to detect legal misclassification in a retrieval based EU AI Act high risk compliance agent, how often does a faithful but wrong answer arise, and what does this imply for how such agents should be evaluated.

2. State of the Art

This work sits at the meeting point of three lines of research. The aim of this section is not to list everything in each line, but to show where each one helps and where it stops, so that the gap this case study fills becomes clear.

Retrieval evaluation and faithfulness metrics. The first line is the work on how to evaluate retrieval based systems. As these systems spread, the community settled on automated metrics that score an answer without a human in the loop. Faithfulness is the most trusted of these, and the RAGAS framework is the common reference implementation. Faithfulness works by breaking an answer into separate claims and checking how many are supported by the retrieved passages. This is exactly the property that makes it useful for spotting hallucination, and exactly the property that makes it blind to legal correctness: it scores grounding, never truth. This case study takes that metric as its subject.

Language models for legal and regulatory classification. The second line applies language models to legal text, including classification tasks where a system or a document must be sorted into a legal category. This work shows the models are capable but also that they make confident mistakes on edge cases, which is precisely where the AI Act's Article 6(3) exception lives. What this line does less of is separating whether the model got the legal answer right from whether its stated reasoning was sound. That separation is the heart of the present work.

AI governance frameworks as context. The third line is the wider governance landscape, in particular the NIST AI Risk Management Framework and the OECD AI work. These frameworks describe how to think about AI risk in general, and they are named in the project brief. They do not, however, provide labelled high risk decisions, so they cannot serve as an answer key for this task. They are used here in two honest ways: as comparative framing that situates the AI Act among other governance approaches, and, in the case of the OECD use case material, as a source of realistic system descriptions to turn into test inputs. All correctness labels come only from the European Commission guidelines.

Reading the three lines together, the position of this work is straightforward. Retrieval evaluation gives us a trusted metric that measures grounding but not legal truth. The legal classification work shows that models fail exactly on the edge cases where grounding and truth can diverge. The governance frameworks give context but no ground truth. Nobody has put the trusted metric on trial by asking, on a task with fixed legal ground truth, whether, and how often, a faithful answer is still legally wrong. That is the question this case study answers.

3. Expected Outcome

The concrete deliverable is a labelled evaluation of one compliance agent, run across up to 150 test inputs, in which every answer carries two independent results: whether its legal label is correct, and how faithful it is to the retrieved text. Sorting the answers by these two results produces four groups, which I refer to as the four buckets: right and faithful, right and unfaithful, wrong and faithful, wrong and unfaithful.

The expected finding is that the two off diagonal groups are not empty. I expect to find real cases in the wrong but faithful group, where the agent's answer is well grounded in the guidelines yet reaches the wrong legal classification, and cases in the right but unfaithful group, where the agent reaches the correct label on reasoning that the retrieved text does not really support. If the wrong but faithful group is not empty, that is the headline result, because it is the case the standard metric is supposed to catch and cannot.

From this I expect to report, in plain numbers, how often the faithfulness metric and the legal ground truth disagree, and to characterise the kinds of inputs where any disagreement clusters, which I expect to be around the Article 6(3) exception. The wider takeaway is a direct answer to the brief's own questions: which parts of this compliance task can be automated, how reliable the automated assessments are, and what this implies for the role of human oversight in the legal correctness check even when the automated grounding score looks reassuring.

4. Proposed Method and Architecture

The system has one agent and one evaluation harness. The agent is the thing being tested. The harness is where the contribution lives. The figure below shows the full flow, from a single test input on the left to the four bucket outcome at the bottom.

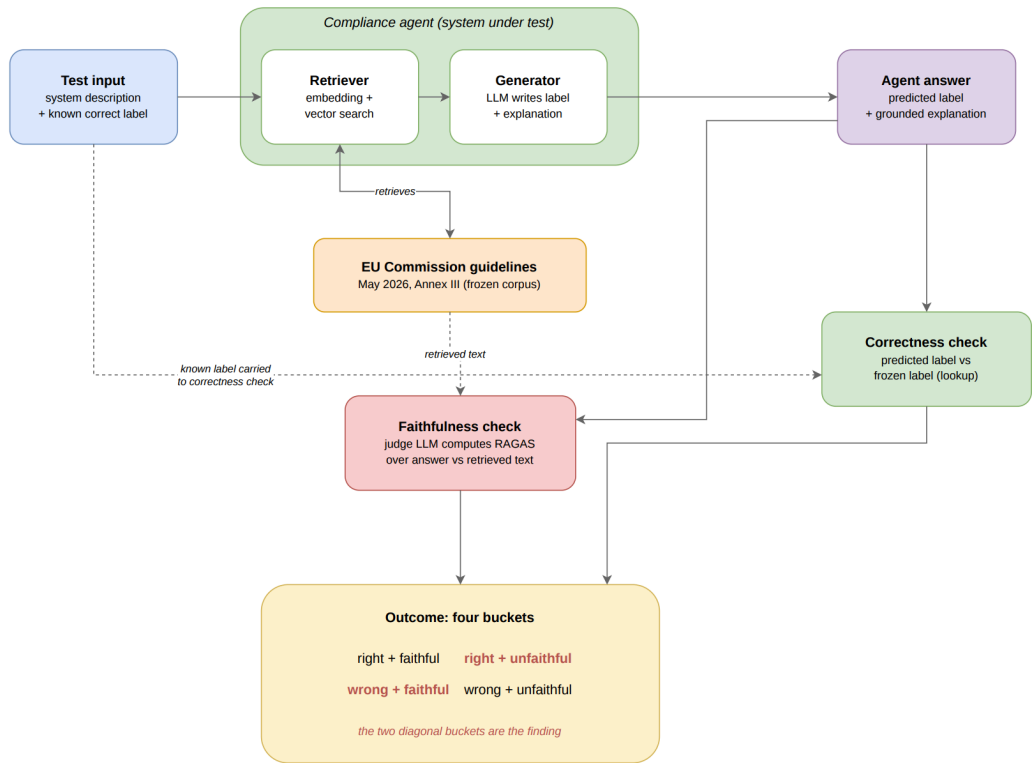


Figure 1. The compliance agent as system under test, evaluated on two independent axes.

A single test input is a description of an AI system paired with its correct high risk label, which is fixed in advance and never shown to the agent. The input goes to the agent, which has two parts. The retriever, an embedding model with a vector search, finds the passages in the frozen Commission guidelines that are most relevant to the description. The generator, a language model, reads those passages and the description and writes an answer: a predicted label together with an explanation grounded in the retrieved passages.

The answer then goes through two checks that run independently of each other. The correctness check is a plain comparison of the predicted label against the fixed label that travelled in with the input. It uses no model and makes no judgment, so it is a lookup. The faithfulness check is computed by a separate judge language model using the RAGAS faithfulness metric, which reads the answer against the retrieved passages and scores how well grounded it is. The judge is deliberately a different model from the generator, so that the answer is not graded by the same model that produced it.

Every answer therefore ends up with two independent results, and each answer falls into one of the four buckets. The two off diagonal buckets, wrong but faithful and right but unfaithful, are the ones where the two axes disagree, and they are the object of the study.

Reused infrastructure. The retrieval and generation pipeline is a RAG orchestrator reused from a prior project and adapted to this domain by changing the corpus and the prompts. It is infrastructure, not a contribution of this work.

Tools and technologies. A Python retrieval and generation stack for the agent, an embedding model with a vector store for retrieval, a frozen copy of the Commission May 2026 Annex III guidelines as the corpus, the RAGAS library for the faithfulness computation, and two distinct language models, one as the generator and one as the judge. The specific generator and judge models are to be fixed before the method is finalised.

5. Evaluation

The evaluation rests on one idea: correctness and faithfulness are two different questions about the same answer, and they must be measured separately so that they are free to disagree. Collapsing them into one number is exactly the mistake this work is built to expose.

Correctness. Correctness asks whether the predicted high risk label matches the true label. The true labels come only from the worked examples in the European Commission's May 2026 draft guidelines on high risk classification, specifically the Annex III material, which are frozen at that version for the whole project. Because the true label is known for every input, correctness is a direct comparison with no judgment involved. This axis answers whether the agent got the law right.

Faithfulness. Faithfulness asks whether the answer is grounded in the retrieved passages, and it is measured with RAGAS faithfulness, the field standard. This is the metric on trial. It is chosen precisely because it is widely trusted and because, by construction, it never sees the true label, so it cannot account

for legal correctness. This axis answers whether the agent's reasoning is grounded, which is a different question from whether it is right.

The two axes are crossed to place every answer in one of four buckets: right and faithful, right and unfaithful, wrong and faithful, wrong and unfaithful. The primary evaluation criterion is the size and content of the two off diagonal buckets. A non empty wrong but faithful bucket is direct evidence that a high faithfulness score does not guarantee a correct legal answer. A non empty right but unfaithful bucket shows the converse, that a correct answer does not guarantee trustworthy reasoning. Alongside the counts, I will inspect a sample of the off diagonal cases by hand to describe why the metric and the truth diverge, with attention to the Article 6(3) exception.

The test set is up to 150 system descriptions, each with a fixed correct label. The descriptions are drawn in part from OECD use case material and are written to include deliberate Article 6(3) edge cases, since those are where the two axes are most likely to separate.

Honest limitations. Two limits are stated up front rather than hidden. First, the justification quality judgments on the off diagonal cases are planned with a single annotator, which is myself. A second annotator, giving an inter annotator agreement figure, would strengthen those judgments and will be added if a labelling partner is available, but the study is designed to stand without one. Second, the faithfulness judge is a language model, so its score is really that model's opinion about how well grounded the answer is, not a hard fact. This does not weaken the argument. It actually helps it. The point of the study is that even when RAGAS is doing exactly what it is meant to do, and giving an answer a good faithfulness score, the answer can still be legally wrong. The metric misses legal errors on its own terms, not because the judge is broken.

6. Timeline

The plan runs across the module window from early July to the end of September 2026, in weekly checkpoints. Submission is targeted about two weeks before the hard deadline to leave room for the colloquium. The Gantt chart below gives the overview, showing how the phases overlap, and the table underneath gives the detailed checkpoint for each week. The dates are anchored to a first week of July start.

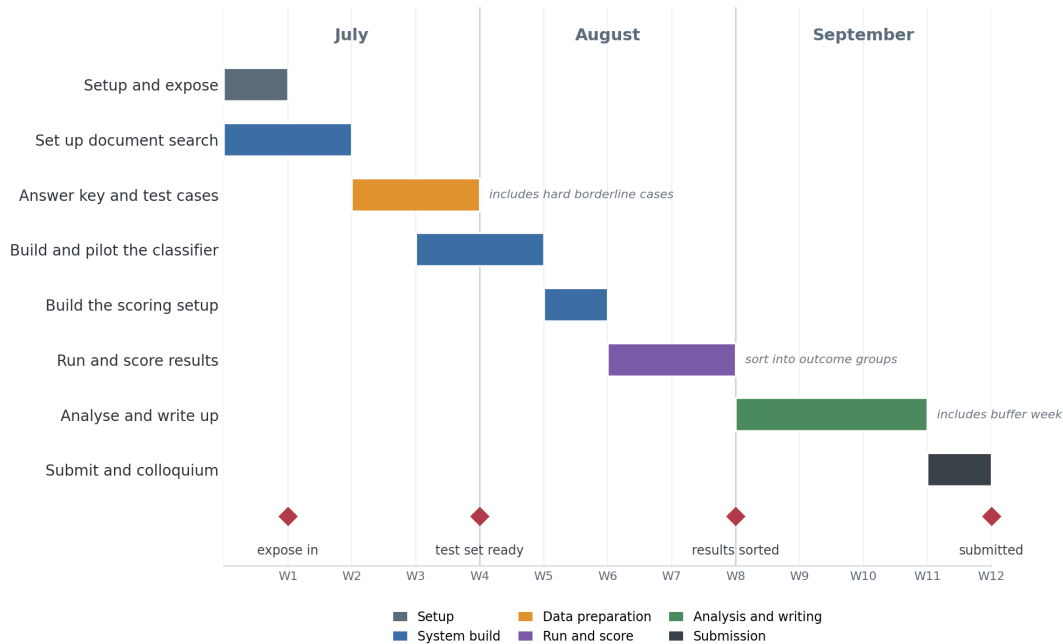


Figure 2. Phase overview across the module. Diamonds mark the main milestones. The table below gives the week by week detail.

Week	Starts	Focus	Checkpoint
Week 1	11 Jul	Finalise and submit exposé. Stand up the reused orchestrator against the new corpus.	Exposé Approval
Week 2	14 Jul	Ingest and chunk the frozen Commission May 2026 guidelines. Get retrieval reliably pulling the right Annex III passages.	Retrieval working
Week 3	21 Jul	Extract the worked examples into a labelled ground truth set. Fix the label schema and the Article 6(3) edge case definitions.	Ground truth frozen
Week 4	28 Jul	Build the test inputs (target 120 to 150 system descriptions) including OECD derived cases and deliberate 6(3) edge cases. Fix one correct label per input.	Test set built
Week 5	4 Aug	Tune the generator prompt to return a label plus a grounded explanation. Pilot run on about 20 inputs and sanity check outputs.	Pilot run done
Week 6	11 Aug	Build the evaluation harness: correctness lookup, plus RAGAS faithfulness with a separate judge model. Fix generator and judge models.	Harness ready
Week 7	18 Aug	Full run of the agent across all test inputs. Collect answers and retrieved passages.	Full run complete
Week 8	25 Aug	Compute both axes. Assign every answer to one of the four buckets. Inspect the off diagonal cases by hand.	Buckets assigned
Week 9	1 Sep	Analysis: how often faithful but wrong and right but unfaithful occur, and where they cluster. Draft the findings.	Findings drafted
Week 10	8 Sep	Optional second annotator sample if available. Robustness checks. Write the discussion, including the human oversight point.	Discussion drafted
Week 11	15 Sep	Write and polish the final report. Buffer for slippage, sickness, and the unexpected.	Draft report done
Week 12	22 Sep	Submit about two weeks before the hard deadline. Prepare the colloquium.	Submitted

7. Bibliography

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